

VM-Based Cloudlets: Giving a Boost to Mobile Computing

Our mobile phones are great at sharing and processing information but do have limitations in terms of storage capacity and processing power. VM-based cloudlets address these drawbacks, leading to seamless mobile computing.



graphics, running machine learning models, or processing large datasets, the computation is offloaded to the VM within the cloudlet. The cloudlet processes the task and sends the results back to the mobile device. This approach reduces latency and conserves the mobile device's resources.

For instance, consider an augmented reality (AR) application used by a mobile device. The device can offload AR rendering tasks to a nearby cloudlet. The VM in the cloudlet processes these tasks efficiently, providing a smoother and more responsive user experience.

Emerging need for cloud computing

The increasing reliance on mobile computing has highlighted the limitations of mobile devices. With the proliferation of applications like augmented reality, real-time gaming, IoT, and edge AI, the demand for low-latency and high-performance computing is growing exponentially. Traditional cloud computing often fails to meet these demands due to high latency and bandwidth constraints caused by the physical distance between users and centralised data centres. VM-based cloudlets address these challenges by offering localised, scalable, and resource-efficient solutions, enabling mobile devices to offload computationally intensive tasks to nearby virtual machines.

As mobile networks expand and the digital ecosystem evolves, the need for VM cloudlets is becoming critical. They empower industries

Mobile computing has transformed the way people access, process, and share information. However, mobile devices often face significant constraints, such as limited battery life, processing power, and storage capacity. VM-based cloudlets have emerged as a solution to bridge the gap between mobile devices and remote cloud services, enabling seamless and efficient mobile computing.

What are VM-based cloudlets?

Cloudlets are small-scale, decentralised data centres located closer to end users.

They are designed to provide low-latency and high-bandwidth computational services. VM-based cloudlets rely on virtual machines (VMs) to deliver computational resources. These VMs encapsulate specific applications, enabling mobile devices to offload resource-intensive tasks to a nearby cloudlet rather than a distant centralised cloud.

How they work

VM-based cloudlets operate by hosting VMs in a localised data centre environment. When a mobile device needs to perform a complex computation, such as rendering